

Getting to School: Survey Data

AP Statistics · Unit 1 · Topic 1.2 · B: 2026-09-10 · E: 2026-09-11 · TEACHER KEY

CED TETHER

- **SKILL: 2.A** — Describe data presented numerically or graphically.
- **ENDURING UNDERSTANDING: VAR-1** Given that variation may be random or not, conclusions are uncertain.
- **VAR-1.B** Identify variables in a set of data. [Skill 2.A]
- **VAR-1.B.1** A variable is a characteristic that changes from one individual to another.
- **VAR-1.C** Classify types of variables. [Skill 2.A]
- **VAR-1.C.1** A categorical variable takes on values that are category names or group labels.
- **VAR-1.C.2** A quantitative variable is one that takes on numerical values for a measured or counted quantity.

Essential question: When is a number a quantity, and when is it just a label?

DOK-3 skill: 4.B · **FRQ pattern:** number-coded-variable-and-meaningless-summary

DOK rationale: Part (c) asks the student to adjudicate two students' classifications of number-coded variables, justify each with the definition, and predict what specific calculation becomes meaningless under the wrong choice; both classifications of the rating can be defended, so the reasoning, not the label, is what is scored.

First take → rules + (a)–(c) → turn in

DOK: 1 → 3

Minutes: 45

Teacher does

- Board slide up at the bell. Read Maya's and Jordan's lines aloud; do not react.
- At minute 5, read the rules box; have students start (a).
- Board slide back up at minute 33. Circulate; push for the definition, not the label.
- Collect at the door. Score part (c) only, E/P/I. Either position on Rating can earn E.

Students do

- One-sentence first take on the zip-code question before any teaching.
- Use the definitions in the rules box to identify individuals and classify the variables.
- Finish (a)–(c); exit line; turn in.

Questions to ask

- What does a zip code measure? What does it count?
- If I averaged two zip codes, what place would I get?
- Do equal one-star steps represent equal changes in what is being measured? State your assumption.
- Which column would you sort to find the 'biggest'? Which column has no 'biggest'?

Adult role

Solo teacher. Circulate 5 pairs. Never say 'categorical' or 'quantitative' yourself during the finish window; ask what the number measures.

[WARN] LOOK FOR

- ‘It has digits, so it’s quantitative’ — confuses how a value is written with what it measures
- Saying zip code is categorical ‘because you can’t add them’ without saying WHY (it measures/counts nothing)
- Treating Rating as obviously quantitative because the quiz app computes an average star rating
- Naming the individuals as ‘the newspaper’ or ‘the school’ instead of the six students

SCORING PART (c) — E / P / I

Expected elements

- **zip-label** (required) — Supports Jordan on ZIP and distinguishes location labels from quantities
- **meaningless-average** (required) — Explains what an average ZIP code fails to measure or describe
- **rating-assumption** (required) — Defends rating as opinion labels, or explicitly assumes an equal-step measured score

E	All three elements are correct and linked to this problem. Accept equivalent plain-language reasoning.
P	At least one element includes a correct evidence-to-reason link, but another is missing or incorrect.
I	No correct evidence-to-reason link; only a choice, copied numbers, or unsupported statements.

Common mistakes

- ‘It has digits, so it’s quantitative’ — confuses how a value is written with what it measures
- Saying zip code is categorical ‘because you can’t add them’ without saying WHY (it measures/counts nothing)
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Optional #1 — not collected

DOK 2 For each variable, write **C** (categorical) or **Q** (quantitative); for each Q, give the unit: (1) jersey number of a soccer player; (2) number of goals scored this season; (3) shoe size; (4) preferred position (goalkeeper, defender, midfielder, forward); (5) minutes played in the last match.

Key: (1) C — a label, even though numeric. (2) Q, goals. (3) Q, shoe-size units (a measured foot length on a scale); accept C with a reason that sizes are ordered labels. (4) C. (5) Q, minutes.

[STAR] TODAY'S PROBLEM — annotated

A school newspaper surveyed six students for a story about getting to school. The table records lunch period, ZIP code, commute time in minutes, and a 1–5-star opinion rating of the commute.

Two reporters disagree about the numbers in it. **Maya:** “Zip code and Rating are both quantitative — they’re numbers, so we can average them.” **Jordan:** “They’re both categorical. Numbers can be labels.”

Getting-to-school survey (minutes; 1–5 stars)

Name	Lunch	ZIP	Min	Stars
Aiden	A	01902	12	4
Bea	B	01905	35	2
Cruz	A	01902	8	5
Dev	C	01904	22	3
Elise	B	01905	41	1
Fiona	A	01901	15	4

FIRST TAKE (5 min)

Would averaging two ZIP codes tell a reader anything useful about getting to school? Give one reason.

Key: Not graded. Most will say ‘yes, it’s a number’; that is the hook.

Rules callout “VARIABLES (keep on the page)” is printed on the student sheet.

DOK 1 (a) Who are the individuals in this data set?

Key: The six surveyed students are the individuals.

DOK 2 (b) Classify Lunch, ZIP, and Min as categorical or quantitative. Give the unit for Min.

Key: Lunch and ZIP are categorical labels; Min is quantitative commute time, measured in minutes.

DOK 3 (c) Decide whose ZIP-code claim is right and explain why an average ZIP code would or would not be meaningful. Then defend a classification for the star rating using what the numbers represent.

Key: Jordan is right about ZIP: the digits label places, not measured or counted quantities. Averaging 01902 and 01905 gives 01903.5, which does not describe an average commute or location. The star rating is categorical when its numbers label ordered opinions. Accept a quantitative interpretation only if the student explicitly assumes equal steps on a measured score scale; writing ‘stars’ alone is not enough.

EXIT

One thing the rules box changed about my first take: _____